

STORM TIME SEASONAL VARIATION OF TEC IN THE SOUTHERN HEMISPHERE MID-LATITUDE REGIONS USING SIGNALS FROM GPS SATELLITE

E. Yizengaw¹ and E. A. Essex

Cooperative Research Center for Satellite Systems, Physics Department, La Trobe University, Australia

¹Fax +61 3 9479 1552; E-mail: e.yizengaw@latrobe.edu.au

ABSTRACT

Total Electron Content (TEC) of the ionosphere over the Australian region has been derived from measurements obtained at GPS ground stations of the International GPS Service (IGS). The derived TEC values at mid- and low-latitudes during the severe magnetic storms of 6 April, 15 July, and 6 November 2000 (equinox, winter, and summer, respectively, for the Southern Hemisphere), show large deviations from the quiet day behaviour. Both an enhancement and a depletion of TEC relative to quiet day are observed during the aforementioned storms and are called positive and negative storm effects, respectively. The observations, in general, indicate that during the magnetic storm the negative disturbance was shifting to middle latitudes from higher latitudes, even to lower latitudes in summer. An enhanced plasma loss over 24 hours has been observed after the storm began at mid-latitudes. Comparisons of GPS TEC with f_oF_2 data from a wide area network of ionosondes in the Australian region show good agreement with a correlation coefficient greater than 0.83.

Keywords: Ionosphere, Magnetic storm, TEC, and GPS

INTRODUCTION AND METHODS

During an ionospheric storm there is heating of the lower part of the thermosphere in the auroral region. The main source of this heating is the joule dissipation of electric currents (Förster & Jakowski, 2000; Danilov and Lastovica, 2001) and it should lead to a significant decrease in the atom-to-molecule ratio throughout the entire high latitude atmosphere. (Buondanto, 1999). Furthermore, this heating causes changes to atmospheric circulation, driving an equatorward wind (Danilov and Lastovica, 2001 and the reference there in). In the simplest approximation, this leads to an equatorward expansion of the negative storm effect (Förster & Jakowski, 2000; Basu et al., 2001). A convenient way of describing the changes is to determine $DTEC\%$, the percentage deviation of TEC from the quiet time value (Förster & Jakowski, 2000).

TEC can be derived from transionospheric GPS signals recorded at the ground because the ionospheric effect on such signals is proportional to the integrated free electron density along the transionospheric ray path. The algorithm used here to calculate TEC from GPS measurements is described by Sardón et al. (1994). Pre-processing of the GPS data is performed to detect and remove or correct signal outliers and signal cycle slips. The derived TEC data are also corrected for receiver and satellite (instrument) biases. The slant TEC data are then mapped to the vertical by applying a mapping function, which is based on a single layer approximation

at 400 km height. All the plots shown have been obtained by combining vertical TEC observations from all 28 GPS satellites that were operating during the campaign.

The objective of this paper is to study the seasonal behaviour of magnetic storms in the Southern Hemisphere, which is less well understood than the Northern Hemisphere response. Data obtained at the IGS stations in the Australian region (see Figure 1) have been used to determine the ionospheric response to magnetic storms occurring in different seasons of the year 2000. The magnetic storms we considered were those of 6 April, 15 July, and 6 November 2000, which occurred during equinox, southern winter and summer, respectively. Latitudinal variation of these storms is considered using the chain of stations at about 150°E meridian. The longitudinal behaviour of the storms is the subject for another paper. All coordinates are given in geographic coordinate system.

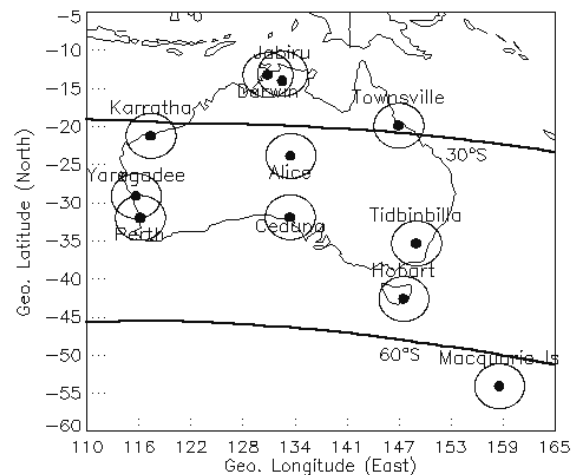


Figure 1: Geographic map of ground based GPS receiver stations in the Australian region. The circles around each station depict 20° elevation coverage. The horizontal curve lines depicts the 30°S and 60°S geomagnetic latitudes.

OBSERVATIONS

April 6 Storm

Following a Coronal Mass Ejection (CME) on 4 April 2000, enhancements in the auroral electrojet (AE) index, exceeding 3000 nT, were recorded at about 18 UT on April 6 as shown in Figure 2a. The corresponding Dst index showed a decrease to -91 nT from -71 nT over a two hour period (between 19 UT and 21 UT) and then decreased further, reaching -321 nT at 01 UT on April 7.

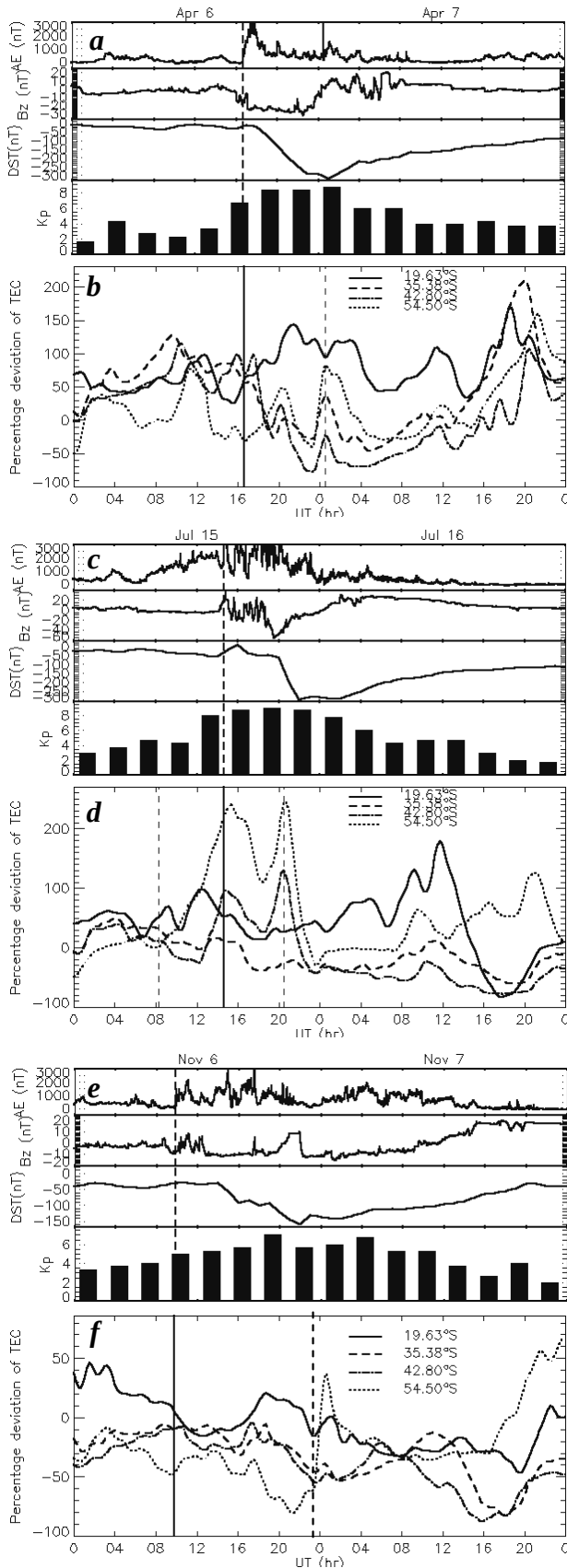


Figure 2. The storm time variation of AE, Dst, B_z (GSM component taken from ACE), and Kp index shown in the panels ((a), (c) & (e)). The panels (b), (d) & (f) depict the corresponding TEC response for April 6, July 15, and November 6, 2000 storm. The dashed lines in (a), (c), & (e) and the solid lines in (b), (d), & (f) indicate the onset time of the storms. The dashed lines at the right in (b), (d) and (f) mark TEC positive response times for the corresponding storm event; and the dashed line at the left in (d) marks the AE-index commencement time.

During the April 6 magnetic storm, the effects of ionospheric ionization irregularities were evident in the GPS data. The negative storm effect was dominant over mid-latitude, and most significantly at 42.80°S latitude, where it lasted for more than 20 hours after the SSC (at 1639 UT). The positive responses were observed at all latitudes before the SSC and purely positive disturbance ($\Delta\text{TEC}\%$ never be negative) at low low-latitudes (e.g. at 19.63°S) as shown in Figure 2b. On April 7 the nighttime TEC enhancement was observed at all stations in the chain..

July 15 Storm

The 15 July 2000 storm was a severe magnetic storm, probably related to the CME emitted from the Sun on July 14 (giving rise to the name "Bastille Day Storm"). As depicted in Figure 2c, the 15 July storm was characterized by a pronounced SSC that started at 1437 UT as indicated by the dashed line. At the height of the storm, the Kp-index achieved its highest possible value of 9. The corresponding Dst index indicates that its magnitude increased rapidly from -20 nT to 10 nT in an hour time gap (between 15 UT and 16 UT) on 15 July. After that it began to decrease, and at 22 UT reached the value -300 nT. On the other hand, the AE-index enhancement reached a value of more than 1500 nT before the SSC.

Hours before the SSC (at 1437 UT), the $\Delta\text{TEC}\%$ over high-latitudes, shown in Figure 2d, revealed a huge positive response, of more than 200%, and after the SSC its magnitude increased further, reaching more than 250% at 2030 UT on July 15. Long lasting positive disturbance was observed at low latitudes (e.g. at 19.63°S) during the storm main phase and recovery phase until it changed its sign at 15 UT on July 16. Like the equinox magnetic storm, the negative storm disturbance was again dominant at mid-latitudes during storm main and recovery phases. During nighttime, between 15 UT and 21 UT on July 16 as shown in Figure 2d, this mid-latitude negative disturbance extended down to lower latitudes. At the same time ionization enhancement occurred over high-latitude regions (e.g. at 54.5°S).

November 6 Storm

The November 6 storm, with $\text{Dst} < -155$ nT and $\text{Kp} > 7$ shown in Figure 2e, also caused long lasting ionization depletion over high- and mid-latitudes, especially at mid-latitudes where the negative storm effect remained for more than 48 hrs (see Figure 2f). Unlike the equinox and winter storm effects, the summer storm negative impact on ionization density extended down to low-latitudes. As can be seen in Figure 2f, the negative storm effect was the dominant effect at 19.63°S after SSC (0947 UT), and mainly during the storm recovery phase.

DISCUSSION AND CONCLUSION

We studied three major magnetic storms by determining the TEC from mid-latitude IGS stations in the Australian region. The onset Storm Time (0 ST) for the three magnetic storm events were during nighttime (pre-

midnight local time for storm on 6 November and post-midnight local time for storms on 6 April and 15 July). We found some similarities and some differences in ionospheric behaviour during the three major magnetic storms in different seasons of the year 2000. All three storms had a similar impact over mid-latitude regions: depleted ionization (negative storm effect) lasting more than 24 hours in winter and summer and for about 20 hours in equinox, which was independent of local time. Such a local-time- independent, negative storm effect over mid-latitudes indicates that the ionization budget was controlled by the storm induced perturbation process. Otherwise, as indicated by theoretical (Fuller-Rowell et al., 1994) and observational (Balan and Rao, 1990) studies, the local time influence on storm pattern cannot be ignored.

According to the current morphology, it is believed that the storm negative disturbance is caused by the change in the thermospheric composition because of the heating of the thermosphere during magnetic storms. The occurrence of storm negative phase at mid- and low-latitudes, shown in Figure 2b, 2d, and 2f, was due to equatorward shift of the molecularly enriched air to lower latitudes by storm induced equatorward directed circulation (referred above). The half-hour averaged storm pattern, in Figure 3, clearly indicates that the storm-induced equatorward directed neutral wind was pushing the negative phase equatorward, after 0 ST of both storm events.

The equatorward penetration of negative phases displayed a seasonal behaviour as shown in Figure 2 and 3. In summer the negative phase was better developed at lower latitudes than in the winter and equinox. Because in summer both dynamic background thermospheric circulation and storm induced circulations reinforce each other and are directed equatorward throughout the day (Danilov and Lastovica, 2001). This might have driven the molecular enriched air further so that it reached the low-latitude regions and caused the negative storm effect over

19.63°S latitudes, which lasted for more than 24 hours (see Figure 2f).

In winter, however, the background and storm induced circulation are not directed in the same direction throughout the day. During nighttime, both the background circulation and the storm-induced winds reinforce each other and are directed equatorward. This might have driven the depleted atom-to-molecule ratio to even lower latitudes and imposed sharp ionization depletions (negative storm effect) such as that observed at 19.63°S (see Figure 2d) during nighttime on July 16. However, in the winter daytime ionosphere, the zone of the depleted atom-to-molecule ratio was restricted to high and mid latitudes as the storm induced equatorward circulation was weakened by background circulation, which is poleward during winter daytime.

The storm positive phase in TEC has been also prominent during these magnetic storms, more significantly at low-latitudes in equinox and winter, which is probably due to the downwelling of thermospheric gas. The downwelling of thermospheric gas, mainly caused by storm induced thermospheric winds (Danilov and Lastovica, 2001), may cause the increase of ionization at low-latitudes without any significant change in atom-to-molecule ratio (see (Fuller-Rowell et al., 1994). Figure 2b & 2d clearly revealed the effect of downwelling of thermospheric gas at low-latitudes, where the downwelling process is stronger, during storm main and recovery phase. In summer, however, only a transient positive phase occurred during the storm main and recovery phase.

The nighttime TEC enhancement on April 7 (see Figure 2b) might be the result of upward drift of plasma along the magnetic field lines, due to equatorward winds (Förster and Jakowski, 2000), to higher altitudes where the recombination rate is low. Because of the magnetic field inclination at mid latitudes (Buondanto, 1999), the nighttime positive disturbance was dominant at mid latitudes (e.g. 35.38°S) as shown in Figure 2b.

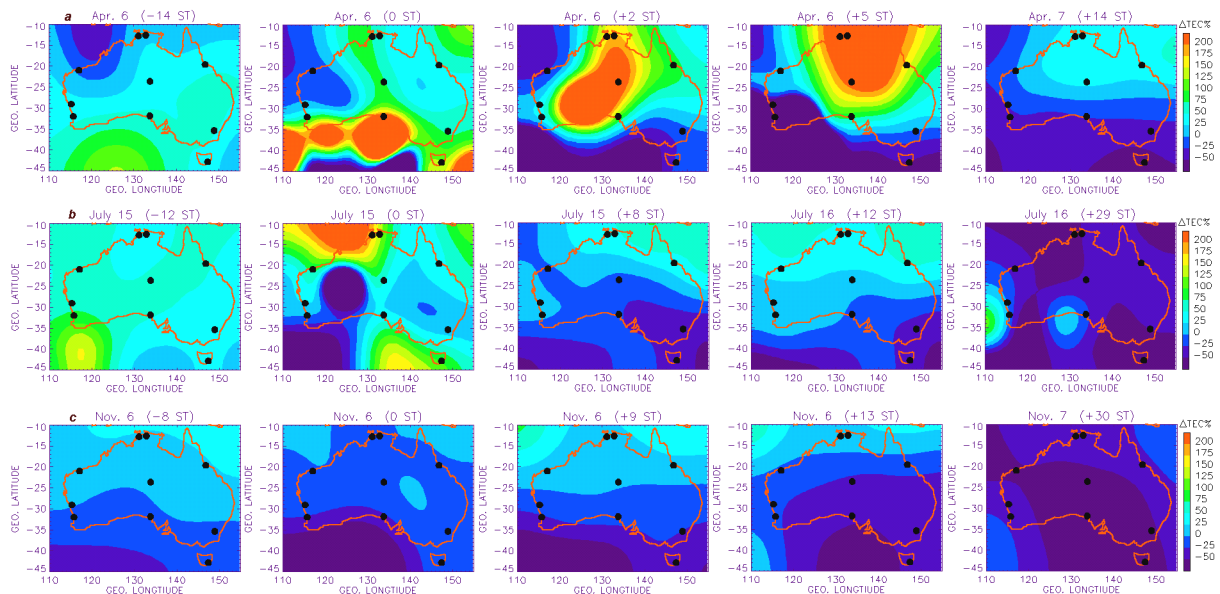


Figure 3. Maps of the averaged TEC response for magnetic storm on (a) April 6, (b) July 15, and (c) November 6, 2000. The black dots depict the IGS stations used for this study.

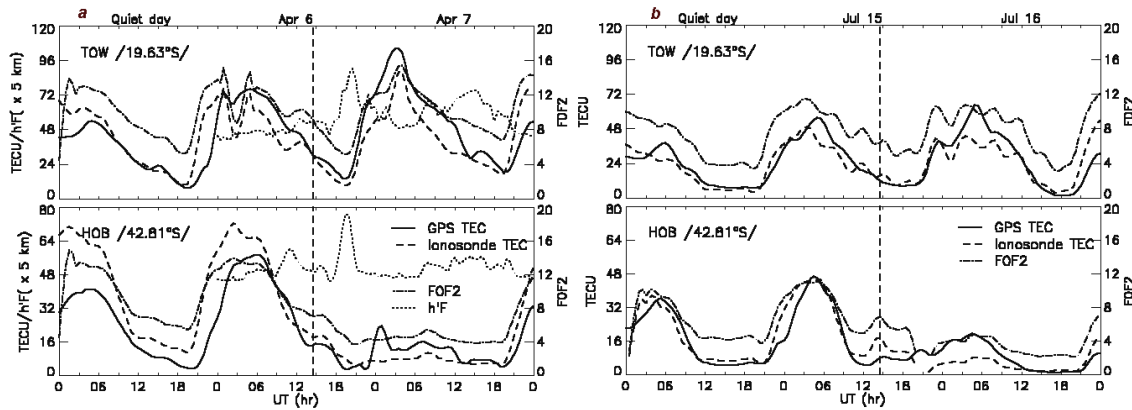


Figure 4. Comparison of GPS TEC with ionosonde data during (a) April and (b) July 2000 storm events. The F-region virtual height ($h'F$) over Townsville and Hobart during April storm is depicted in (a).

The earlier (before 0 ST) enhancement of AE-index, in Figure 2d, indicated high energy input to the high latitude regions. This implies that the energetic particles might also precipitate (Buonsanto, 1999) to this regions leading to the ionization enhancement at high latitudes (Basu et al., 2001) such as that observed at 54.5°S during storm main phase on July 15 shown in Figure 2d. The simultaneously observed peaks at 0030 UT on April 7, 2030 UT on July 15, and 01 UT on November 7 are assumed to be correlated with an enhanced eastward electric field, which has a direct link with the southward directed IMF Bz-component (Wygant et al., 1998; Basu et al., 2001). The nighttime positive storm disturbances observed at 54.5°S latitude in between 08 UT and 22 UT on July 16 (see Figure 2d) might be the result of the local dynamo electric field initiated by equatorward winds (Danilov and Lastovicka, 2001).

The ionosonde data, f_0F_2 and TEC derived from f_0F_2 using appropriate slabthickness (Huang et al., 2001), also exhibit a similar fluctuation with GPS TEC, as shown in Figures 4a & b. Figure 4a depicts the virtual ionospheric minimum virtual height of the F-layer ($h'F$) sudden enhancement starting about 17 UT on April 6, followed by a decrease in f_0F_2 and GPS TEC, which is a clear indication of spread F occurrence (Buonsanto, 1999). The large increases in $h'F$ levels during post-sunset hours are also an indication of an increase in the eastward electric field (Lakshmi et al., 1991), which is consistent with the above explanation. In conclusion, the analysis of the variation of GPS-based TEC at the chain of GPS stations located at different latitudes permitted us to identify the following overall effects:

- The negative phase equatorward shift during the storm from high to middle latitudes, and even to low latitudes in summer.
- positive storm effect over low latitudes during the storm due to the downwelling of gas.
- penetration of magnetospheric origin electric field to the altitude of ionosphere region and cause a simultaneous TEC peak.

All these perturbations following the magnetic storm, which induced TEC variations over mid- and low-latitude regions, may have adversely affected the performance of navigation and communication systems.

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